Equal Latency, Selective Quality:
Hidden Compute and Evidence Selection in AI Service Markets

Zhenyuan Pan[a,*]

[a] *Graduate Student, Department of Economics,*
Graduate School of Arts and Science, New York University,
New York, NY, USA

Abstract

AI providers can deliver different answer quality at the same price and response time by varying the computation behind each answer. This paper studies how a provider's private prediction of whether an answer will be checked shapes this hidden allocation of quality. In a dynamic model, users purchase before the provider learns request-level costs and the likelihood of verification, and public performance affects future sales. Without audits, the prospect of future sales encourages the provider to favor requests that are more likely to be verified. When effort differs across requests, publicly recorded accuracy therefore exceeds the average accuracy delivered to buyers, even though buyers understand the provider's incentives. Audits targeting requests that lack public evidence can weaken or reverse this quality ordering. Numerical examples show that auditing can induce uniformly high effort or shift effort toward less visible requests. Separately, random outcome checks reduce bias in the public record when delivered quality and service incentives are held fixed. With known sampling probabilities and a complete request list, partial verification also permits unbiased quality estimates. The analysis highlights two distinct tasks for verification policy: improving the quality users receive and making measured performance representative.

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*Corresponding author: Zhenyuan Pan.
Email address: zp2411@nyu.edu (Zhenyuan Pan)

1. Introduction

A premium reasoning service sells quality that is difficult to observe at purchase. A buyer can compare its price and promised waiting time with a faster alternative. The buyer may later discover whether the answer was correct. Productive computation, however, takes place inside the provider's system. This creates an economic question for AI service markets: what determines which purchased requests receive the costly effort that the premium service promises?

Answer release can separate the visible wait from productive effort. Suppose compute x determines completion time $h(x)$, while the provider can add post-completion padding $d \geq 0$. The buyer observes

$$T = h(x) + d. \tag{1}$$

A high-compute answer with little padding and a low-compute answer with more padding can share this display. The same service label and release commitment can therefore conceal different expected accuracy across requests.

Market discipline depends on what this observable display can reveal. Signaling models relate the credibility of an observable choice to its cost of imitation (Spence, 1973; Kihlstrom and Riordan, 1984; Milgrom and Roberts, 1986). Release control gives a lower-compute action a way to reproduce the waiting-time signal while saving resources. Reputation offers another source of discipline: public performance can put future rents at risk (Klein and Leffler, 1981; Shapiro, 1983; Fudenberg and Levine, 1992). Competition and market participation help determine the value of those rents (Hörner, 2002; Board and Meyer-ter Vehn, 2013), while platform feedback can connect performance to future allocation (Tadelis, 2016). This raises a further question about which requests generate the evidence that disciplines effort.

We study a provider that privately predicts future verification after the purchase. Users choose a premium Deep service or a competitive Fast alternative using the equilibrium quality schedule. After Deep adoption, the provider learns an incremental compute cost and a profile that predicts natural public evidence. High compute improves correctness and the prospect of favorable evidence, which protects future selling opportunities under an external exposure rule. The private profile consequently changes the return to effort on a request that the user has already purchased. A common release time conceals the resulting allocation.

The mechanism builds on the economics of difficult-to-assess quality. Quality uncertainty can impede exchange (Akerlof, 1970), and experience or credence attributes govern what buyers can learn through use (Nelson, 1970; Darby and Karni, 1973). Expert-service models study private information about service needs (Wolinsky, 1993; Emons, 1997; Dulleck and Kerschbamer, 2006), with second opinions and verifiability changing the incentives to supply appropriate service (Pesendorfer and Wolinsky, 2003; Fong, 2005; Dulleck et al., 2011; Fong et al., 2014). Our provider's additional information concerns whether performance will become verifiable. Its forecast affects the quality delivered before the evidence is realized.

Evidence selection links this hidden allocation to measured performance. Research on selective labels shows how observed outcomes can cover a selected set of decisions (Lakkaraju et al., 2017; Kleinberg et al., 2018); self-selected reviews raise a related issue for consumer ratings (Li and Hitt, 2008). Here the provider can change service quality using a private forecast of inclusion. The quality schedule and the subsequent evidence sample are jointly endogenous. Users know the equilibrium schedule, so the public-sample gap measures the difference between recorded and delivered quality. The external exposure rule supplies the intertemporal reward for public performance.

Verification can affect the service decision or the information it leaves behind. Contract theory ties incentives to observable signals (Holmström, 1979; Grossman and Hart, 1983). Disclosure and certification research studies how information about quality reaches the market (Grossman, 1981; Milgrom, 1981; Lizzeri, 1999; Albano and Lizzeri, 2001; Dranove and Jin, 2010). Recent industrial organization work examines platform disclosure (Gambato and Peitz, 2025), business models and quality investment (Allain et al., 2025), and service quality under consumer search (Moraga-González and Sun, 2025). We distinguish two verification instruments within this broader problem. Effort auditing changes the return to hidden compute; random outcome attestation changes the coverage of the public record.

AI contracting gives these observability questions a specific production setting. Algorithmic contract theory studies rewards for unobserved actions (Bernasconi et al., 2024; Dütting et al., 2024), and statistical contracts apply incentive design to generated text (Saig et al., 2024). Hidden technology and effort choices enter LLM delegation in Sreenivas and Larson (2026); Saig et al. (2026) study costly evaluation chosen after an initial signal. Our focus is the provider's private forecast of verification after a user commits to purchase. This information links the hidden action to the composition of later evidence.

Payment and verification rules identify different hidden-compute margins. Artola Velasco et al. (2026a) study test-time compute incentives under usage-based pricing and a procurement mechanism. At a fixed price, our provider instead weighs request-level cost savings against future exposure. Token-reporting incentives and their sequential auditing concern billed quantities (Artola Velasco et al., 2026c,b). Model-substitution tests (Cai et al., 2025) and hidden-token diagnostics (Sun et al., 2025a,b; Wang et al., 2025) target other aspects of unobserved operation. Verifiable computation offers a way to check execution (Guo et al., 2026). In the model, an audit's value comes from its ability to distinguish productive actions, summarized by the gap between their signal probabilities.

The central result connects the provider's incentive slope to public-sample distortion. For a common cost distribution, recorded quality exceeds or falls below adopter quality according to the weak sign of the covariance between evidence posteriors and the cutoff policy. The incentive slope determines that sign. With uniform costs and interior actions, the magnitude equals the slope times adopter-weighted posterior variance, scaled by the accuracy gain and natural evidence coverage. Missing-evidence audits act on the slope. Independent outcome attestation acts on inclusion at a fixed quality schedule, while known sampling probabilities permit unbiased estimation from a partially observed request roster.

The equilibrium illustrations show how these incentives interact with adoption. The baseline has a zero branch and two positive roots of the exposure-value map. Its upper attracting root under conjecture iteration supports substantial compute differentiation at a common latency. Missing-evidence auditing eventually supplies high compute to every request in the baseline. A wider cost support permits interior sorting reversal on an equilibrium branch. Price and exposure paths also reach local folds, where the positive branches meet as future trade becomes insufficient to sustain nearby high-quality outcomes.

The analysis gives verification policy two distinct economic tasks. It can change which requests receive costly service, and it can change which delivered outcomes enter the record. Private forecasts of verifiability connect these tasks because the provider can tailor effort to expected visibility. Tracing this connection through adoption and future exposure explains why similar-looking services can leave a systematically selective public record.

The rest of the paper proceeds as follows. Section 2 presents the market environment. Section 3 derives the hidden-action cutoff and dynamic equi-

librium. Section 4 establishes the analytical results. Section 5 reports the numerical illustration. Section 6 discusses market design and scope. Section 7 concludes.

2. Model

2.1. Services and timing

The provider trades current compute costs against future selling opportunities. It serves short-lived users through an external marketplace over an infinite sequence of discrete periods. The provider operates Deep, while the marketplace commits to an exposure rule that the provider cannot revise. An independent auditor supplies audit signals and collects fines. A period is one service-opportunity cycle, discounted by $\delta \in (0, 1)$.

Public exposure determines how often the provider can make a sale. The state is $s \in \{H, L\}$, and a representative Deep opportunity arrives with probability $n_s \in [0, 1]$, where $n_H > n_L$. Conditional on an opportunity, the user compares the provider's Deep offer with a competitive Fast alternative. Their posted prices and release times are

$$q_F = (P_F, T_F), \qquad q_D = (P_D, \tau), \tag{2}$$

where $P_D > P_F$ and $\tau > T_F$. Deep therefore asks the user to pay more and wait longer. The provider earns profit from Deep adoption; Fast is supplied by the competitive outside market.

The purchase decision precedes the provider's request-level information. The user first observes the offers and exposure state, then chooses a service using the equilibrium quality schedule. After Deep adoption, the provider learns the private evidence profile z and incremental compute cost c. It chooses productive compute and enough padding to meet τ. Natural outcome evidence or an audit can subsequently update public exposure. This order lets the provider tailor effort after the buyer has committed to the service.

Exposure scales opportunity arrival while preserving the conditional user environment. We study stationary equilibria that choose a common solution of the adoption-posterior system in both states. Within this class, conditional compute policies and adoption are state invariant. Section 3.3 states this restriction formally and derives the corresponding value equation.

2.2. Productive compute and latency camouflage

The same Deep offer can be fulfilled with two compute allocations. Action G uses high productive compute x_H. Action P uses $x_L < x_H$, followed by a longer interval of release padding.

Release control makes the two actions observationally identical in waiting time:

$$h(x_H) + d_G = h(x_L) + d_P = \tau. \tag{3}$$

Since $h(\cdot)$ is strictly increasing,

$$d_P - d_G = h(x_H) - h(x_L) > 0. \tag{4}$$

Users observe the sum of productive completion time and padding. A shorter computation can therefore produce the same waiting-time experience as a longer one. The difference appears in expected correctness.

Productive compute raises the probability of a correct answer. For the binary outcome $Y \in \{0, 1\}$, let

$$\Pr(Y = 1 \mid P) = p_L, \qquad \Pr(Y = 1 \mid G) = p_L + \Delta p, \tag{5}$$

where $0 \leq p_L < p_L + \Delta p \leq 1$. This accuracy gain costs the provider an additional request-level amount c, above the low-compute resource cost C_P. The cost has common continuous distribution H_c on $[0, \bar{c}]$, whose CDF is zero below the support and one above it. The numerical illustration uses uniform costs. Both actions are feasible when $h(x_H) \leq \tau$.

The binary technology isolates the decision to supply the accuracy increment. Section 4.1 gives the corresponding camouflage argument for continuous compute and allows padding to be costly. In the baseline, padding carries no separate flow charge.

2.3. Users and private profiles

Users differ in the value of correctness and the visibility of outcomes. Group i has population mass μ_i, correctness stake w_i and natural evidence probability m_i. Individual value v and waiting cost λ are random. Users know the provider's equilibrium policy, but the realized evidence profile z remains private to the provider.

The private profile supplies the provider's information about future verification. Group masses sum to one and $w_i > 0$. After adoption, the provider observes (z, c); group identity and user preferences remain unobserved. Conditional on group, (v, λ) is independent of z and the natural-evidence indicator.

In the numerical benchmark, value and waiting cost are independent, with $\gamma_i > 0$. The cost draw is independent of group, profile, user preferences and natural evidence, so it adds no information about verification.

Natural verification is independent of the shock that determines correctness. Within group i, evidence arrives with probability m_i, independently of the profile and adoption. Correctness follows Equation (5) conditional on the compute action. These assumptions imply $E \perp Y \mid z, \text{adoption}$, as well as independence within group. Natural evidence is resolved by the end of the cycle, after which the remaining requests can be audited. Given profile distribution $Q(z \mid i)$, the provider's evidence posterior among Deep adopters is

$$m_z = \Pr(E = 1 \mid z, \text{Deep adoption}) = \frac{\sum_i \mu_i d_i Q(z \mid i) m_i}{\sum_i \mu_i d_i Q(z \mid i)}, \tag{6}$$

whenever the denominator is positive. Here d_i is adoption conditional on a Deep opportunity. The weights reflect who buys: a group that values the anticipated service more strongly contributes more requests to the posterior. Credibility therefore changes the information on which the provider acts.

Let

$$g_z = \Pr(G \mid z) \tag{7}$$

be the high-compute probability after profile z. Before adoption, a user averages this policy over the profiles that the provider may observe. We call the resulting probability group credibility:

$$\chi_i = \sum_z Q(z \mid i) g_z. \tag{8}$$

Credibility translates the hidden policy into expected service quality. Deep accuracy for group i is $\widehat{p}_i = p_L + \chi_i \Delta p$, which gives utilities

$$u_D = v w_i \widehat{p}_i - P_D - \lambda \tau, \qquad u_F = v w_i p_L - P_F - \lambda T_F. \tag{9}$$

The outside option has utility zero.

For $\chi_i > 0$, Deep adoption requires the value threshold

$$\bar{v}_i(\lambda; \chi_i, P_D) = \max \left\{ \frac{P_D + \lambda \tau}{w_i (p_L + \chi_i \Delta p)}, \frac{P_D - P_F + \lambda(\tau - T_F)}{w_i \chi_i \Delta p}, 0 \right\}. \tag{10}$$

The user must prefer Deep to both available alternatives. The first term is the threshold against non-purchase; the second is the threshold against Fast. At

$\chi_i = 0$, Deep offers no accuracy gain despite its price and latency premium, so $d_i = 0$.

Demand aggregates these purchase thresholds over user preferences. In the numerical benchmark, v is exponential with rate γ_i and λ is uniform on $[0, \bar{\lambda}_i]$. Conditional Deep demand is

$$d_i = \frac{1}{\bar{\lambda}_i} \int_0^{\bar{\lambda}_i} \exp\left[-\gamma_i \bar{v}_i(\lambda; \chi_i, P_D)\right] d\lambda. \tag{11}$$

Higher credibility lowers the relevant interior cutoffs and attracts more Deep buyers. Opportunity arrival scales this demand to $n_s d_i$ in public state s. Because the same factor n_s multiplies every group, it cancels from the posterior in Equation (6).

2.4. Evidence and audit

Auditing supplies evidence where naturally observed outcomes are missing. A natural signal is favorable when $Y = 1$ and adverse when $Y = 0$. Each remaining request enters a dual-clock audit with probability $a \in [0, 1]$. Its favorable-signal probabilities are α_G under high compute and α_P under low compute, with

$$\Delta_A = \alpha_G - \alpha_P > 0. \tag{12}$$

The detection gap measures how well the audit distinguishes productive actions. Signal probabilities satisfy $0 \leq \alpha_P < \alpha_G \leq 1$. Conditional on the action, the audit signal is independent of group and cost. An adverse signal triggers a fine $F \geq 0$, paid to the independent auditor outside the user transaction. To implement this signal with completion timestamps, trusted execution must bind the timestamps to the stipulated computation.

External transfers can reward compute or naturally verified correctness. A sponsor pays $\zeta_x \geq 0$ through a separately authenticated compute meter on every covered transaction, and $\zeta_Y \geq 0$ when natural evidence records a correct answer. Users continue to pay the posted price. The benchmark sets both rates to zero; retaining them in the analytical cutoff identifies how such contracts affect the same request-level incentive.

High compute improves public evidence through either available verification channel. For profile z, its incremental favorable-signal probability is

$$\Delta e_z = m_z \Delta p + (1 - m_z) a \Delta_A. \tag{13}$$

The first term is the accuracy gain weighted by natural coverage. The second is the audit detection gap weighted by coverage among the remaining requests. Their sum determines how much public exposure the provider puts at risk by saving on compute.

3. Equilibrium discipline and public exposure

3.1. The hidden-action cutoff

The provider supplies high compute when its incremental return covers the cost. Let $\Delta V = V_H - V_L$ denote the continuation-value gain from high exposure. For profile z, the return from transfers and public-signal consequences is

$$\begin{aligned}\mathfrak{D}_z = \zeta_x \Delta x + m_z \zeta_Y \Delta p + (1 - m_z) a \Delta_A F \\ + \delta \left[m_z \Delta p + (1 - m_z) a \Delta_A \right] \Delta V,\end{aligned} \tag{14}$$

where $\Delta x = x_H - x_L$. High compute is optimal exactly when

$$c \leq \mathfrak{D}_z. \tag{15}$$

It follows that

$$g_z = H_c(\mathfrak{D}_z). \tag{16}$$

For uniform costs, $g_z = [\mathfrak{D}_z / \bar{c}]_0^1$, where $[y]_0^1 = \min\{1, \max\{0, y\}\}$.

The cutoff collects the rewards for supplying the accuracy increment. Verified compute earns the input transfer, while naturally observed correctness earns the outcome transfer. High compute also reduces expected audit fines and improves future exposure. The provider compares their sum with c. Integrating this decision over costs yields a probability of high compute that can vary across evidence profiles.

Current profit accounts for the costs of the resulting allocation. The unconditional expected incremental compute cost after profile z is

$$K_z = \int_0^{\min\{\max\{\mathfrak{D}_z, 0\}, \bar{c}\}} c \, dH_c(c). \tag{17}$$

The expected margin from an adopter with profile z is

$$\begin{aligned}M_z = P_D - C_P + \zeta_x \left(x_L + g_z \Delta x \right) + m_z \zeta_Y \left(p_L + g_z \Delta p \right) \\ - K_z - (1 - m_z) a \left[1 - \alpha_P - g_z \Delta_A \right] F.\end{aligned} \tag{18}$$

The margin combines posted revenue with transfers, then subtracts resource costs and expected fines. A change in audit coverage therefore affects current profit as well as the incentive to invest. Both effects enter the value of future exposure.

3.2. Signals and state transitions

Compute choices determine the public signals that move exposure. Profile-level accuracy and audit favorability are

$$A_z = p_L + g_z \Delta p, \qquad R_z = \alpha_P + g_z \Delta_A. \tag{19}$$

For group i and profile z, favorable and adverse public-signal probabilities are

$$q_{iz}^{+} = m_i A_z + (1 - m_i) a R_z, \tag{20}$$

$$q_{iz}^{-} = m_i (1 - A_z) + (1 - m_i) a (1 - R_z). \tag{21}$$

The remaining probability produces no public signal.

The marketplace rewards favorable evidence with higher future opportunity arrival. An adverse signal moves H to L, while a favorable signal moves L to H. A signal in the other direction, or no signal, leaves the state unchanged. The transition probabilities are

$$\ell_H = n_H \sum_i \mu_i d_i \sum_z Q(z \mid i) q_{iz}^{-}, \tag{22}$$

$$u_L = n_L \sum_i \mu_i d_i \sum_z Q(z \mid i) q_{iz}^{+}. \tag{23}$$

An exposure transition requires both a sale and a relevant public signal. The opportunity factor n_s scales the chance of a sale; group demand and profile probabilities determine the signal distribution conditional on that opportunity.

The same adoption weights determine the provider's expected current return. Profit conditional on a Deep opportunity is

$$\bar{\pi} = \sum_i \mu_i d_i \sum_z Q(z \mid i) M_z. \tag{24}$$

Flow profit in state s is $\pi_s = n_s \bar{\pi}$. Provider values satisfy

$$V_H = \pi_H + \delta \left[(1 - \ell_H) V_H + \ell_H V_L \right], \tag{25}$$

$$V_L = \pi_L + \delta \left[u_L V_H + (1 - u_L) V_L \right]. \tag{26}$$

Subtracting Equation (26) from Equation (25) gives

$$\Delta V = \frac{\pi_H - \pi_L}{1 - \delta + \delta(\ell_H + u_L)}. \tag{27}$$

The numerator measures the current profit advantage of high exposure. The denominator discounts that advantage and accounts for the signals that erase the initial state difference. Consequently, a verification change can alter exposure value even when it raises current profit in both states.

3.3. Equilibrium correspondence

Equilibrium requires consistency within the service cycle and across exposure states. For a conjectured value gap V, the cutoff determines allocation, which determines credibility and demand. Demand in turn determines the evidence posterior in Equation (6). Let $\mathcal{M}(V)$ collect the posterior vectors that solve this inner feedback. Each $m \in \mathcal{M}(V)$ generates a Bellman value $\mathcal{H}(V, m)$ through Equation (27).

A common-inner-solution stationary equilibrium is a pair $(\Delta V^*, m^*)$, implemented in both public states, such that

$$m^* \in \mathcal{M}(\Delta V^*), \qquad \Delta V^* = \mathcal{H}(\Delta V^*, m^*). \tag{28}$$

Several internally consistent adopter compositions can exist at the same value conjecture. The equilibrium object is therefore a correspondence. Cancellation of n_s gives both states the same inner equations. The equilibrium class additionally requires them to choose the same solution m^*.

The common solution permits exposure to act through opportunity arrival alone. It supports the factorization $\pi_s = n_s \bar{\pi}$ used in the value bound and fold calculations. The compute cutoff and sample-covariance accounting apply within each public state under the maintained information structure. For two groups, Appendix A.3 gives sufficient conditions for an inner solution at a positive conjecture. These conditions permit multiple solutions.

At zero demand, beliefs must be specified before any buyer is selected. We use the potential-opportunity posterior

$$m_z^0 = \frac{\sum_i \mu_i Q(z \mid i) m_i}{\sum_i \mu_i Q(z \mid i)}. \tag{29}$$

This profile-specific belief defines behavior off the trading path. With positive demand, the posterior weights the users who actually buy. As demand

approaches zero, a small selected set of buyers can dominate those weights, so its limiting composition may differ from the off-path convention.

Every equilibrium in this class has a bounded value gap. Let $\bar{\pi}^{\text{abs}}$ bound the absolute per-opportunity margin. Since conditional demand is at most one,

$$|\Delta V| \leq \bar{B}_V \equiv \frac{|n_H - n_L|\bar{\pi}^{\text{abs}}}{1-\delta}. \tag{30}$$

Suppose the composite map $h_*(V) = \mathcal{H}(V, m(V))$ is continuous on a closed interval, with an inner solution selected at each positive-demand point. Opposite signs of $h_*(V) - V$ at the endpoints then supply a fixed point. Continuity of the composite map suffices even if the posterior itself jumps at a zero-demand endpoint. In the unaudited, zero-transfer benchmark, allocations and demand vanish uniformly as $V \downarrow 0$, so $h_*(V) \to 0$; Appendix A.4 proves this statement. A zero endpoint alone establishes the boundary root and supplies no positive-root existence claim. We describe a root as locally attracting under conjecture iteration when

$$\left|\frac{d\mathcal{H}}{dV}\right| < 1. \tag{31}$$

The attraction criterion refers to the conjecture-adjustment map used in computation.

No trade is self-consistent when exposure has no incentive value. With $a = \zeta_x = \zeta_Y = 0$ and conjecture $\Delta V = 0$, the cutoff is $\mathfrak{D}_z = 0$, so $g_z = 0$ when $H_c(0) = 0$. Deep then offers Fast accuracy at a higher price and a longer wait. Demand and exposure value are both zero. Positive branches can coexist through the reverse feedback: credibility attracts buyers, which makes future market access worth protecting.

4. Economic results

4.1. Equal-latency camouflage

Release control lets different productive actions share one visible service category. The following result states the feasible substitutions and the cost condition under which a provider prefers lower compute.

Proposition 1 (Compute nonidentification from displayed latency)**.** *Fix a posted release time τ. Let productive compute be $x \in [\underline{x}, \overline{x}]$, with strictly increasing completion time $h(x)$. Write $C(x, c)$ for production cost and $\psi(d)$*

for padding cost. Every feasible x satisfying $h(x) \leq \tau$ generates the display $T = \tau$ through $d(x) = \tau - h(x)$. If two such compute choices exist, displayed latency alone cannot identify compute. In the binary dynamic model, let $\Delta C = C(x_H, c) - C(x_L, c)$ and $\Delta\psi = \psi(\tau - h(x_L)) - \psi(\tau - h(x_H))$. A same-display deviation from high to low compute is strictly profitable exactly when

$$\Delta C - \Delta\psi > \mathfrak{D}_z. \tag{32}$$

If padding is infeasible and h is strictly increasing, equal displayed latency identifies productive compute.

The cost saving must exceed the lost reward from public evidence. In the benchmark, $\Delta C = c$ and padding has zero marginal cost, so lower compute is chosen when $c > \mathfrak{D}_z$. This is the complement of the cutoff rule in Equation (15). Costly padding raises the saving required for a same-display deviation. Execution attestation can narrow the feasible substitutions by binding a completion timestamp to the stipulated computation.

4.2. Selective compute provision

The evidence profile changes how much the provider gains from quality. Differentiating the cutoff in Equation (14) with respect to m_z gives the sorting slope

$$B(a, \Delta V) \equiv \frac{\partial \mathfrak{D}_z}{\partial m_z} = \Delta p(\zeta_Y + \delta\Delta V) - a\Delta_A(F + \delta\Delta V). \tag{33}$$

Greater natural verifiability raises the return from observed correctness. It also reduces the chance that the request enters the missing-evidence audit pool. These two effects enter B with opposite signs, which makes audit design relevant to the ordering of service quality.

Proposition 2 (Verifiability sorting)**.** *Fix ΔV and the adoption-selected posteriors. Suppose H_c is strictly increasing at the relevant cutoffs. For two profiles z' and z with $m_{z'} > m_z$,*

$$\operatorname{sign}(g_{z'} - g_z) = \operatorname{sign} B(a, \Delta V) \tag{34}$$

whenever both action probabilities are interior. If $B(a, \Delta V) > 0$, requests with greater natural verifiability receive more productive compute and less padding. If $B(a, \Delta V) < 0$, the direct ordering reverses.

Sorting occurs within the posted category that users have already chosen. Both profiles receive an answer at τ, although the provider can give them different probabilities of the accuracy increment. The observable wait therefore leaves this quality allocation hidden.

Buyer selection changes how informative the two profiles are. Consider equal-mass groups $i \in \{\ell, h\}$, with $m_\ell < m_h$ and conditional demands d_ℓ, d_h. A binary profile predicts group identity with symmetric accuracy $\rho > 1/2$. Among Deep adopters, Bayes' rule gives

$$m_{z_h} - m_{z_\ell} = (m_h - m_\ell)\frac{d_\ell d_h(2\rho - 1)}{[d_\ell(1-\rho) + d_h\rho][d_\ell\rho + d_h(1-\rho)]}. \tag{35}$$

The denominator equals

$$d_\ell d_h + \rho(1-\rho)(d_\ell - d_h)^2. \tag{36}$$

Unequal demand puts the same overrepresented group into both profile pools. The nonnegative square term therefore compresses the posterior gap relative to the equal-demand benchmark $(2\rho - 1)(m_h - m_\ell)$. This feedback limits the differentiation induced by profiling while allowing selective provision to persist.

4.3. The public evidence sample

The quality schedule shapes the record from which outsiders observe performance. Let $E \in \{0, 1\}$ indicate natural verification of a Deep outcome, and let $Y \in \{0, 1\}$ indicate correctness. With $p_E = \Pr(E = 1) > 0$, average correctness in the public sample is

$$S_0 = \mathbb{E}[Y \mid E = 1] = \frac{\mathbb{E}[EY]}{p_E}. \tag{37}$$

Average correctness among all Deep adopters is $\bar{p} = \mathbb{E}[Y]$.

Lemma 1 (Evidence-sample identity)**.** *The public-sample gap satisfies*

$$\boxed{S_0 - \bar{p} = \frac{\mathrm{Cov}(E, Y)}{\Pr(E = 1)}.} \tag{38}$$

For any profile z, define $m(z) = \Pr(E = 1 \mid z)$ and $p(z) = \mathbb{E}[Y \mid z]$. Then

$$\boxed{S_0 - \bar{p} = \frac{\mathrm{Cov}(m(z), p(z)) + \mathbb{E}[\mathrm{Cov}(E, Y \mid z)]}{\mathbb{E}[m(z)]}.} \tag{39}$$

Selection can operate between profiles or within a given profile. Equation (39) separates these sources. In the strategic model, conditional independence sets the within-profile term to zero. The remaining covariance measures whether profiles that appear more often in the record also receive more accurate service. Section 6.2 discusses how outcome-dependent verification changes the incentive equations.

Truthful outcomes can give an unrepresentative picture of delivered quality. If the provider supplies more effort to requests likely to be verified, the record overweights better-served requests. The gap can therefore arise through service allocation and inclusion alone.

Two user groups make this composition effect explicit. Let their adopter shares be $\omega_\ell, \omega_h > 0$, with $m_h > m_\ell$ and symmetric profile accuracy $\rho > 1/2$. Under within-group independence of natural evidence and correctness, $g_{z_h} > g_{z_\ell}$ implies

$$\mathrm{Cov}(E, Y) = \omega_\ell \omega_h (m_h - m_\ell)(2\rho - 1)(g_{z_h} - g_{z_\ell})\Delta p > 0. \tag{40}$$

Each factor has an economic role. Verification differences and an informative profile connect group identity to the provider's action. Selective compute then gives the more visible group higher expected accuracy. Their product determines the positive public-sample gap.

The cutoff turns the sample identity into an incentive result. Write its intercept as $A(a, \Delta V) = \zeta_x \Delta x + a\Delta_A(F + \delta \Delta V)$, so $\mathfrak{D}_z = A + Bm_z$. The relevant distribution is the one among buyers; expectations under this profile distribution are denoted by $\mathbb{E}_A$.

Proposition 3 (Incentives and evidence distortion)**.** *Consider a common-inner-solution stationary equilibrium with positive adoption and $p_E > 0$. Under the model's conditional independence assumptions and a common cost CDF H_c,*

$$S_0 - \bar{p} = \frac{\Delta p}{\mathbb{E}_A[m_z]} \mathrm{Cov}_A(m_z, H_c(A(a, \Delta V) + B(a, \Delta V) m_z)) \,. \tag{41}$$

The gap is nonnegative when $B > 0$, nonpositive when $B < 0$, and zero when $B = 0$. For $B \neq 0$, its sign is strict exactly when the action probability is nonconstant with positive adopter probability. If costs are uniform on $[0, \bar{c}]$ and all actions are interior, the exact magnitude is

$$\boxed{S_0 - \bar{p} = \frac{\Delta p \, B(a, \Delta V)}{\bar{c} \, \mathbb{E}_A[m_z]} \mathrm{Var}_A(m_z).} \tag{42}$$

In this interior case, positive posterior variance gives a strict common sign for the quality ordering and public-sample gap whenever $B \neq 0$.

The formula joins the incentive to differentiate service with the opportunity to do so. The slope B governs the response of quality to verifiability. Posterior variance measures the differences in verifiability across purchased requests. More accurate profiling can change that variance through adoption, which makes buyer composition part of the equilibrium effect. A constant allocation at a cost-support corner also removes bias under the common accuracy technology. The proposition gives a sign relation for general cost densities; the audit-path response additionally depends on how probabilities and adopter weights adjust.

4.4. Random outcome attestation

Random attestation reaches requests that the natural record leaves out. After computation, draw $R \sim \text{Bernoulli}(r)$ independently of (E, Y, z). An external verifier obtains a reliable correctness label for every sampled request, including those without a natural public record. Inclusion in the expanded sample is

$$E_r = E + (1 - E)R. \tag{43}$$

Proposition 4 (Independent random attestation)**.** *Hold the joint distribution of (E, Y, z) fixed. The expanded-sample gap is*

$$\boxed{S_r - \bar{p} = \frac{(1 - r)\,\text{Cov}(E, Y)}{r + (1 - r)p_E}.} \tag{44}$$

If $\text{Cov}(E, Y) \neq 0$, then $|S_r - \bar{p}|$ is strictly decreasing in r and converges to zero as r approaches one.

The proposition isolates the effect of broader outcome coverage. Quality and adoption are fixed, and the added labels leave exposure and transfers unchanged. Post-compute timing hides the realized draw when effort is chosen. If the extra labels entered the exposure rule, their anticipated probability would also affect the provider's incentive.

Known inclusion probabilities support quality estimation at partial coverage. With a roster of N requests and independent draws $R_j \sim \text{Bernoulli}(r)$,

where $r > 0$, the inverse-probability construction of Horvitz and Thompson (1952) gives

$$\widehat{p}_r = \frac{1}{N}\sum_{j=1}^{N}\left[E_j Y_j + \frac{(1-E_j)R_j Y_j}{r}\right]. \tag{45}$$

Each sampled missing outcome represents $1/r$ such requests. Conditional on the roster's (E_j, Y_j), the estimator's expectation over the draws is $N^{-1}\sum_j Y_j$. Thus partial certification permits unbiased estimation while the unweighted pooled record remains selective. Lower coverage raises conditional variance, as shown in Appendix A.9. The designer can distinguish the goal of a representative raw record from the goal of an unbiased quality estimate.

4.5. Audit neutralization and reversal

Missing-evidence audits direct additional discipline toward less visible requests. Differentiating the cutoff with respect to coverage, at a fixed continuation value, gives

$$\frac{\partial \mathfrak{D}_z}{\partial a} = (1-m_z)\Delta_A(F+\delta\Delta V). \tag{46}$$

When $F + \delta\Delta V > 0$, a request receives a higher compute cutoff as audit coverage rises. The factor $1 - m_z$ makes this increase larger for profiles less likely to produce natural evidence. Auditing can therefore raise effort while reducing its dependence on natural visibility.

The same targeting changes the slope of the quality schedule:

$$\frac{\partial B(a,\Delta V)}{\partial a} = -\Delta_A(F+\delta\Delta V). \tag{47}$$

The zero of this slope determines which profile has the higher cutoff.

Proposition 5 (Missing-evidence audit and slope neutralization)**.** *Fix ΔV and adoption. Suppose $\Delta_A(F + \delta\Delta V) > 0$. The signed verifiability slope is zero at*

$$a^\dagger = \frac{\Delta p(\zeta_Y + \delta\Delta V)}{\Delta_A(F+\delta\Delta V)}. \tag{48}$$

When $a^\dagger \in (0,1)$, high-verifiability profiles receive the larger direct high-compute incentive for $a < a^\dagger$. The profile slope is zero at $a = a^\dagger$. Low-verifiability profiles receive the larger direct incentive for $a > a^\dagger$, provided action probabilities remain interior.

Better discrimination between actions lowers the coverage needed to neutralize the slope. A larger Δ_A reduces the threshold for $B = 0$. Action probabilities can also become equal when both cutoffs reach the same cost-support corner. The numerical analysis therefore distinguishes interior neutralization from full high-compute provision.

The full equilibrium response includes the change in future rents. Along a differentiable common-solution branch,

$$\frac{dB(a, \Delta V(a))}{da} = -\Delta_A(F + \delta\Delta V) + \delta(\Delta p - a\Delta_A)\frac{d\Delta V}{da}. \tag{49}$$

The first term retains the direct coverage effect. The second transmits the endogenous change in exposure value. Demand also changes the distribution of adopter posteriors, so the sample gap can adjust through both incentives and composition.

The cost density determines how cutoff changes translate into allocation changes. Hold ΔV and $m_{z_h} > m_{z_\ell}$ fixed. For density f_c, the derivative of $G(a) = g_{z_h} - g_{z_\ell}$ is

$$\begin{aligned} G'(a) = \Delta_A(F + \delta\Delta V)\big[&(1 - m_{z_h})f_c(\mathfrak{D}_{z_h}) \\ &- (1 - m_{z_\ell})f_c(\mathfrak{D}_{z_\ell})\big]. \end{aligned} \tag{50}$$

Uniform density gives equal weight to equal cutoff changes, so the interior derivative is negative. A general density can put more mass near the high-profile cutoff. For example, let $H_c(c) = c^3$ on $[0, 1]$, $(m_{z_\ell}, m_{z_h}) = (0.2, 0.8)$, $\delta\Delta V = 1$, $\Delta p = 0.4$, $\Delta_A = 0.5$, and zero fines and transfers. At $a = 0$, the cutoffs are 0.08 and 0.32, with $G'(0) = 0.02304 > 0$. The shrinking cutoff gap then accompanies an initially increasing probability gap.

The choice of audit pool determines whether direct sorting can reverse. Consider an independent audit draw for every request. Whenever an audit occurs, its signal determines exposure; otherwise natural evidence determines exposure. Natural-outcome transfers remain payable whenever natural evidence exists. Under this signal-priority rule, the discipline index is

$$\begin{aligned} \mathfrak{D}_z^{\text{all}} = \zeta_x\Delta x + m_z\zeta_Y\Delta p + a\Delta_A F \\ + \delta[(1 - a)m_z\Delta p + a\Delta_A]\Delta V. \end{aligned} \tag{51}$$

Universal coverage supplies the same audit incentive to every profile. Its verifiability slope is $\Delta p[\zeta_Y + (1 - a)\delta\Delta V]$, which is nonnegative for $\Delta V \geq 0$.

This rule can flatten the gradient while preserving its weak ordering. The baseline reversal comes from the extra audit reward directed toward requests missing natural evidence.

Equal coverage parameters imply different numbers of audits. At fixed behavior, missing-evidence auditing examines a share $a(1 - p_E)$ of adopters, compared with a under universal auditing. At equal audit volume, the universal rule instead uses coverage $a(1 - p_E)$; its slope remains nonnegative for $\Delta V \geq 0$. The sorting comparison thus survives this volume adjustment under the specified signal priority.

4.6. Pricing and exposure boundaries

Price affects discipline through both margins and future market size. Raising P_D increases revenue per adopter while reducing the number of buyers attracted by Deep. The lost adoption can eventually outweigh the higher margin, at which point nearby positive-equilibrium branches may meet at a fold.

Fix a C^2 local selection $m(V, P_D)$ of the inner correspondence near a candidate fold. Such a selection follows from the implicit function theorem when the inner residual is C^2 and its posterior Jacobian is nonsingular. The cutoffs must remain on smooth pieces. Define

$$F^{\mathrm{fp}}(V, P_D) = V - \mathcal{H}(V, m(V, P_D); P_D). \tag{52}$$

A local saddle-node boundary satisfies

$$F^{\mathrm{fp}} = 0, \qquad F^{\mathrm{fp}}_V = 0, \qquad F^{\mathrm{fp}}_{VV} \neq 0, \qquad F^{\mathrm{fp}}_{P_D} \neq 0. \tag{53}$$

The conditions identify a local meeting of two fixed-point branches. On one side of the critical premium, the two nearby roots disappear. The same calculation applies to the low-state opportunity probability n_L. As n_L approaches n_H, losing high exposure costs fewer future opportunities, which weakens the incentive to supply high compute.

Hidden quality depends on the trade that future exposure can preserve. The premium must leave sufficient adoption for high exposure to be valuable. The exposure rule must also create a sufficiently large opportunity difference across states. These restrictions connect the request-level cutoff to market participation.

5. Numerical illustration

5.1. Parameterization and method

The numerical illustration follows the feedback from hidden effort to market participation. Higher credibility attracts Deep buyers, whose composition changes the provider's evidence posteriors. Those posteriors determine compute choices and public-state transitions. Equation (28) closes this loop by requiring the resulting exposure value to match the provider's conjecture.

Table 1 specifies the theoretical economy used throughout the illustration. The two groups have equal masses and correctness stakes, so their evidence probabilities supply the initial source of differentiation. The symbol ρ denotes the accuracy of the provider's private profile.

Table 1: Baseline parameterization

Parameter	Value	Economic role
$p_L, \Delta p$	0.45, 0.25	Low-compute correctness and high-compute gain
$\gamma, \bar{\lambda}$	2.00, 0.20	Value-distribution rate and waiting-cost support
T_F, τ	0.00, 0.40	Fast and Deep displayed latency
P_F, P_D	0.35, 0.38	Fast and Deep prices
n_H, n_L	1.00, 0.55	Deep-opportunity probabilities
$\delta, \bar{c}$	0.90, 0.04	Discount factor and incremental-cost support
C_P	0.02	Low-compute resource cost
μ_ℓ, μ_h	0.50, 0.50	User-group masses
m_ℓ, m_h	0.20, 0.80	Natural evidence probabilities
w_ℓ, w_h	1.00, 1.00	Correctness stakes
ρ	0.75	Symmetric profile accuracy
a	0.00	Baseline audit coverage
α_G, α_P	0.80, 0.20	Favorable audit probabilities
F	0.01	Fine after an adverse audit signal
ζ_x, ζ_Y	0.00, 0.00	Contingent transfer rates
$x_L, \Delta x$	0.20, 0.40	Compute units used in allocation summaries
β_h	0.50	Slope in the summary relation $h(x) = \beta_h x$

The computation resolves demand before searching for consistent exposure values. Demand is integrated analytically between intersections of the affine value cutoffs and checked against a separate trapezoid calculation. Log-domain posterior ratios preserve composition when adoption is small. The bound in Equation (30) determines the outer search interval. We scan it on a mixed-resolution grid, which has spacing at most 10^{-5} below $V = 0.01$ and at most 5×10^{-4} thereafter. Brent refinement uses an absolute tolerance of 10^{-13} on ordinary brackets; representable near-zero candidates receive relative checks.

Each average describes a different part of the market. Population credibility is $\bar{\chi}_\mu = \sum_i \mu_i \chi_i$, while adopter credibility is $\bar{\chi}_A = \sum_i \mu_i d_i \chi_i / \sum_i \mu_i d_i$. The latter determines delivered accuracy, $\bar{p} = p_L + \Delta p \bar{\chi}_A$. Natural-sample quality S_0 additionally weights requests by their chance of appearing in the public record. Profit per opportunity is $\bar{\pi}$; multiplying by n_s gives state profit. Stationary average profit uses the invariant distribution over public states.

5.2. Baseline equilibrium branches

The baseline supports both no trade and positive Deep provision. Its value-gap mapping has three nonnegative roots, reported in Table 2. Their local attraction or repulsion describes numerical conjecture iteration.

Table 2: Baseline roots of the value-gap mapping

Root	ΔV	Conjecture-iteration classification
1	0.0000	Attracting boundary
2	0.0362	Repelling interior root
3	0.2150	Attracting positive root

The upper positive branch supports substantial differentiation within the Deep service. At this root, population-weighted credibility is 0.6193 and adopter-weighted credibility is 0.6241. High-compute probabilities are 0.4383 for the low evidence profile and 0.8003 for the high evidence profile. Thus users face a common release time while the provider allocates high compute with probabilities that differ by 0.3621.

Future exposure can sustain the quality on which its value depends. Figure 1 shows this consistency condition as both a map and a residual. At zero credibility, the Deep menu is dominated and attracts no buyers. The middle root separates local basins under conjecture iteration. At the upper root,

credibility attracts enough adoption to make the loss of future exposure costly, which supports continued investment in compute.

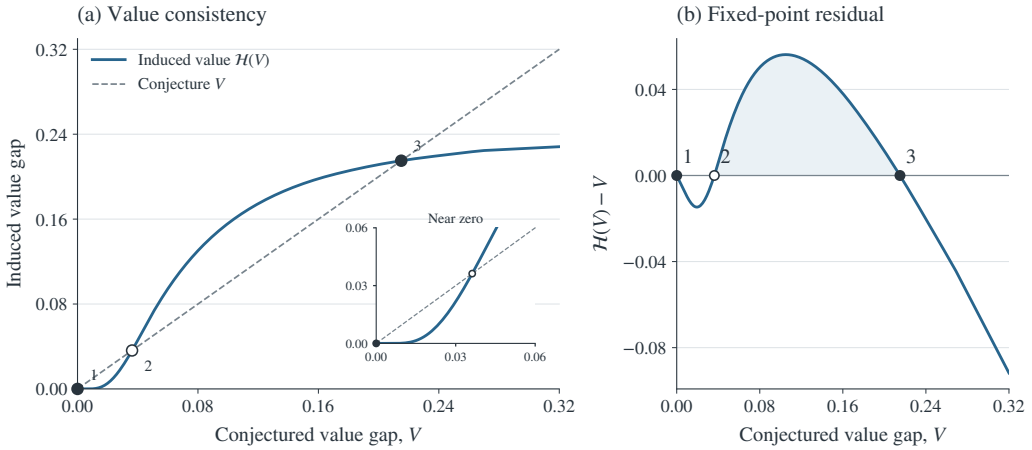


Figure 1: Baseline exposure-value consistency. Panel (a) shows the induced-value map, with a near-zero enlargement; panel (b) shows its residual. Numbered roots match Table 2. Filled markers attract under conjecture iteration; the open marker repels.

Independent calculations recover the reported baseline quantities. Root locations agree to the displayed precision across three search grids. A separate implementation compares the selected value gap and allocations, along with the implied transition probabilities and Bellman values; its maximum absolute difference is below 10^{-10}. Six posterior starts check initialization sensitivity. Exact demand differs from trapezoid integration by less than 10^{-8}.

5.3. Evidence-sample distortion

The public record gives greater weight to the better-served group. Average correctness is 0.6060 among all Deep adopters and 0.6192 in the natural evidence sample. The difference is 0.0131, or 1.31 percentage points. The composition bars in Figure 2 show why: the high-verifiability group supplies 52.67 percent of adopters and 81.66 percent of naturally observed outcomes. Lemma 1 expresses this selection effect as a covariance.

Random outcome coverage brings the record closer to delivered quality. Table 3 applies Equation (44) at the baseline quality schedule and adopter distribution. Figure 2(c) places the same six coverage rates beside the

composition comparison. Increasing coverage monotonically reduces the gap, which reaches zero when every outcome is observed.

Table 3: Evidence-sample gap under random outcome attestation

Attestation rate r	Evidence-sample gap
0.00	0.01312
0.10	0.01079
0.25	0.00797
0.50	0.00447
0.75	0.00193
1.00	0.00000

Attestation changes inclusion after productive effort has been chosen. As in Proposition 4, labels are reliable and sampling leaves the baseline allocation and adopter weights fixed. The falling gap therefore measures the coverage channel directly.

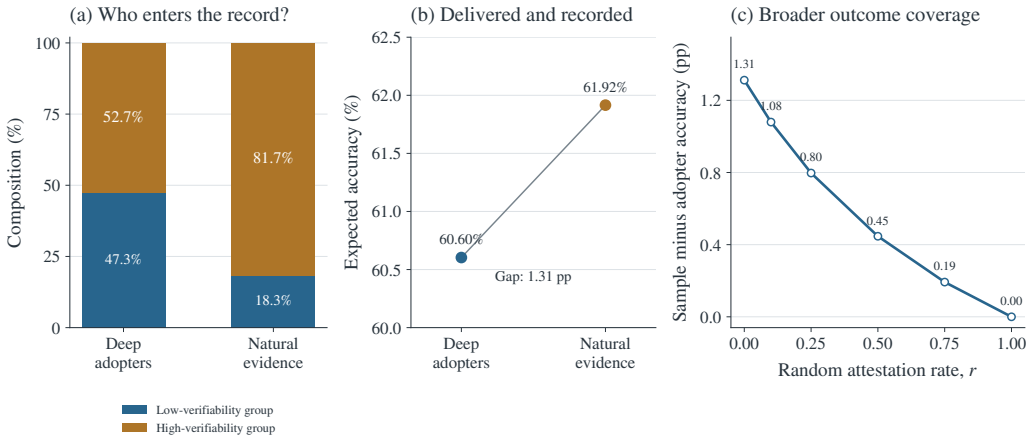


Figure 2: From delivered service to the public record. Panel (a) compares adopter and natural-evidence composition; panel (b) compares their expected accuracy. Panel (c) plots the six fixed-schedule attestation cases in Table 3. Accuracy differences are in percentage points (pp).

5.4. Audit coverage and profiling

Missing-evidence audits expand productive effort on the upper attracting branch. Figure 3 follows the highest positive root that attracts under conjecture iteration. As coverage rises from zero to 0.45, average credibility increases from 0.6193 to one and the score-level allocation gap closes. Conditional population-weighted adoption rises from 0.2496 to 0.3019; profit per opportunity increases from 0.0877 to 0.1025. The value gap falls from 0.2150 to 0.1987, so stronger current effort coexists with a smaller continuation-value difference.

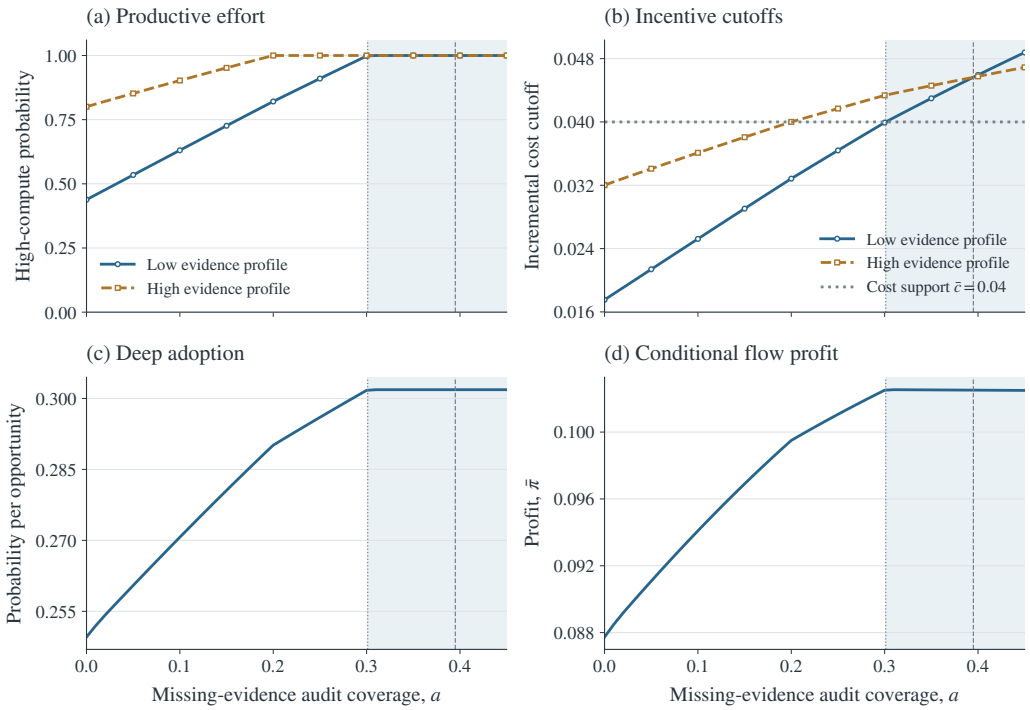


Figure 3: Audit coverage on the upper attracting branch. Panels show allocation, incentive cutoffs, adoption and profit per opportunity. The dotted guide marks joint saturation at $a = 0.30118$; the dashed guide marks $B = 0$ at $a = 0.39503$. Shading identifies the region with full high-compute provision.

The cost boundary explains why the baseline allocation gap closes. Both action probabilities reach one at $a = 0.30118$, where $\min_z \mathfrak{D}_z = \bar{c}$ and $B > 0$. The cutoff slope reaches zero later, at $a = 0.39503$. By then both cutoffs

exceed the cost support. Thus the baseline path displays saturation; interior sorting reversal requires room for action probabilities to move within their support.

A wider cost support allows the ordering to reverse within equilibrium. Figure 4 sets $\bar{c} = 0.05$ and keeps the other baseline primitives fixed over the same coverage interval $[0, 0.45]$. Demand, posteriors and continuation values adjust jointly. At $a = 0.45$, the selected root has $\Delta V = 0.1934124$, with action probabilities $g_{z_\ell} = 0.9508297$ and $g_{z_h} = 0.9137417$. Both are interior. The lower-evidence profile now receives more compute, which gives the natural sample a negative quality gap. The separate gap panels distinguish this reversal from the saturation observed at $\bar{c} = 0.04$; the value panel shows all detected roots alongside the selected branch.

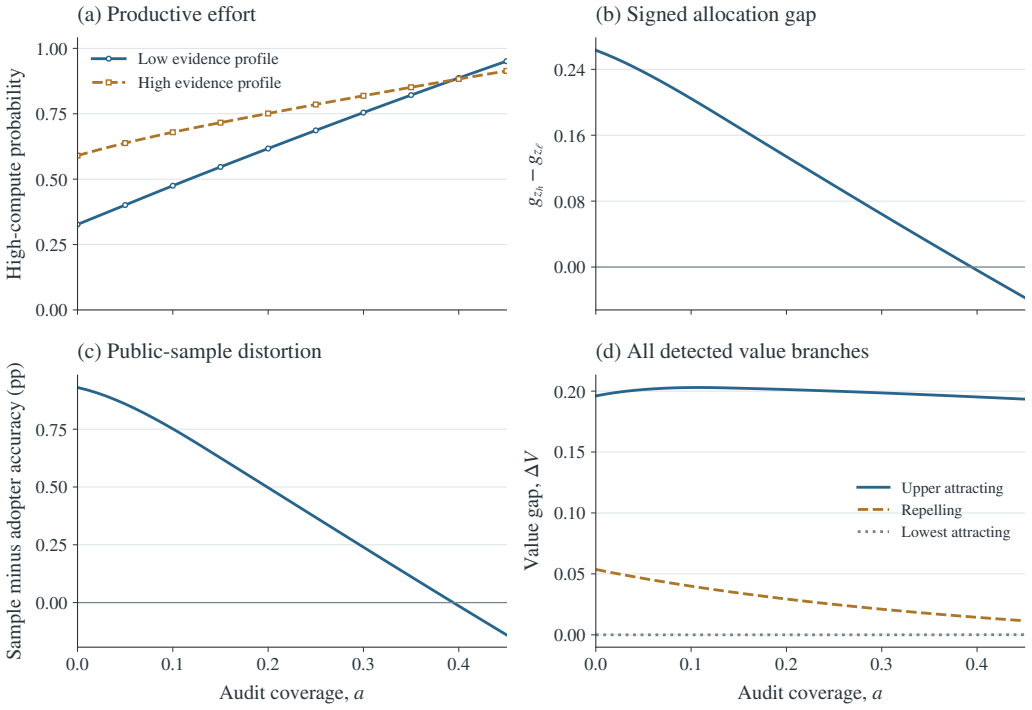


Figure 4: Equilibrium sorting reversal with $\bar{c} = 0.05$. Panels (a)–(c) follow the highest detected positive root attracting under conjecture iteration: action probabilities, their signed gap and the natural-sample accuracy gap in percentage points. Panel (d) includes all 138 detected roots across the 46 coverage points.

More accurate profiling separates service quality without monotonically raising its average. In Figure 5, population credibility starts at 0.6116 when $\rho = 0.5$, rises above that level at the baseline $\rho = 0.75$, then falls to 0.5868 at $\rho = 0.9$. The score allocation gap grows from zero to 0.5344. Conditional population-weighted adoption falls between the endpoints from 0.2490 to 0.2324. The comparison of population and adopter averages makes the accompanying selection visible: profiling changes who receives effort and who finds Deep worth buying.

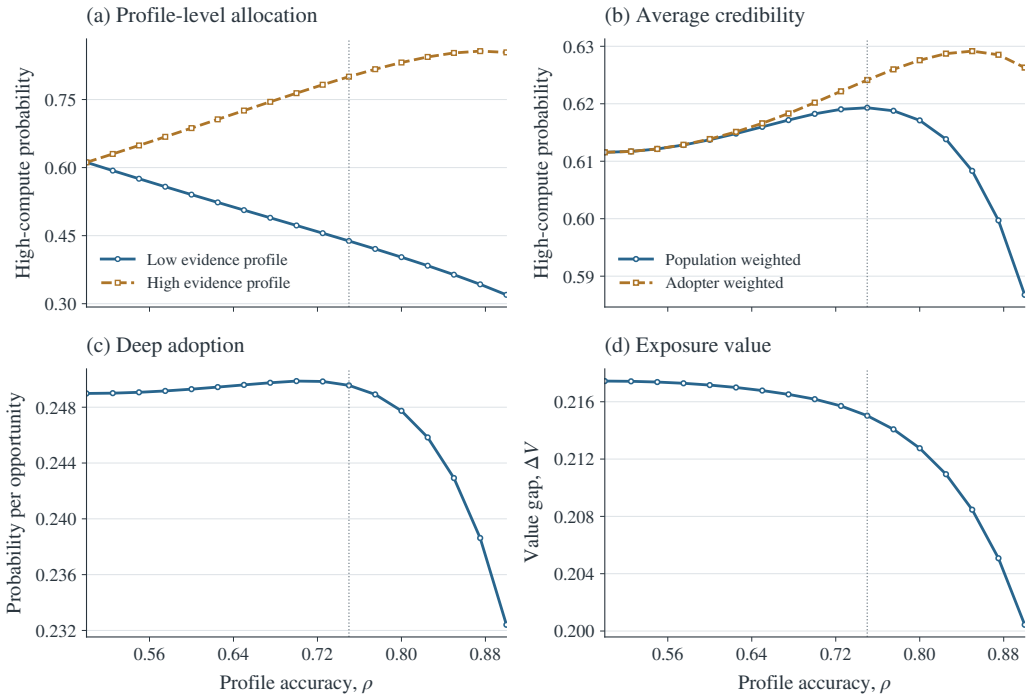


Figure 5: Profile accuracy on the selected locally attracting branch. Panels report profile-level allocation, population/adopter credibility, adoption and exposure value. Adopter credibility weights group allocation by $\mu_i d_i$. The dotted guide marks the baseline $\rho = 0.75$; all 17 path points are shown.

5.5. Premium and exposure folds

The premium faces a limit through its effect on adoption. Along the price path, the positive branches meet at the local fold

$$P_D^{\text{crit}} = 0.4430, \qquad \Delta V^{\text{crit}} = 0.1702. \tag{54}$$

The derivative values are $F^{\mathrm{fp}}_{VV} = 9.4425$ and $F^{\mathrm{fp}}_{P_D} = 0.8748$, with finite-difference step checks and a nonsingular inner posterior Jacobian supporting the local calculation. The implied premium over Fast is $P_D - P_F = 0.0930$. Figure 6 shows the full branch relation beside an enlargement of the meeting point.

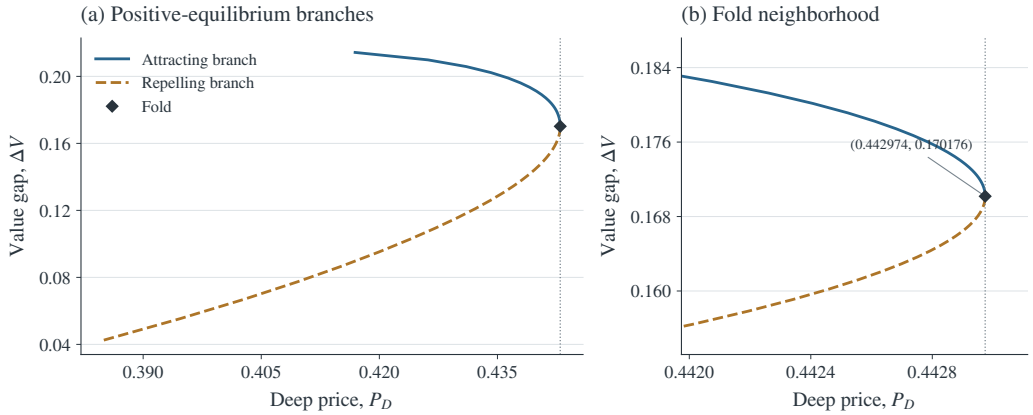


Figure 6: Deep-price branches and their fold. The right panel enlarges the marked neighborhood. Solid and dashed curves indicate attraction and repulsion under conjecture iteration; the diamond is the solution of the fold equations.

Margin per buyer cannot sustain quality without sufficient future trade. A higher price initially supports the provider's return per adopter. Beyond the local fold, the contraction in adoption leaves too little exposure value to support the nearby positive branches.

The exposure rule determines how much future trade is at stake. As the low-state opportunity probability rises, the positive branches collide at

$$n_L^{\mathrm{crit}} = 0.7154, \qquad n_H - n_L^{\mathrm{crit}} = 0.2846, \qquad \Delta V^{\mathrm{crit}} = 0.0735. \tag{55}$$

The two public states become economically similar as n_L approaches n_H. Losing the high state then costs the provider fewer selling opportunities. Figure 7 traces the resulting loss of discipline through the branch relation.

5.6. Evidence heterogeneity

Differences in natural verification create the scope for selective service. Equal evidence probabilities give a zero score-level allocation gap. Moving

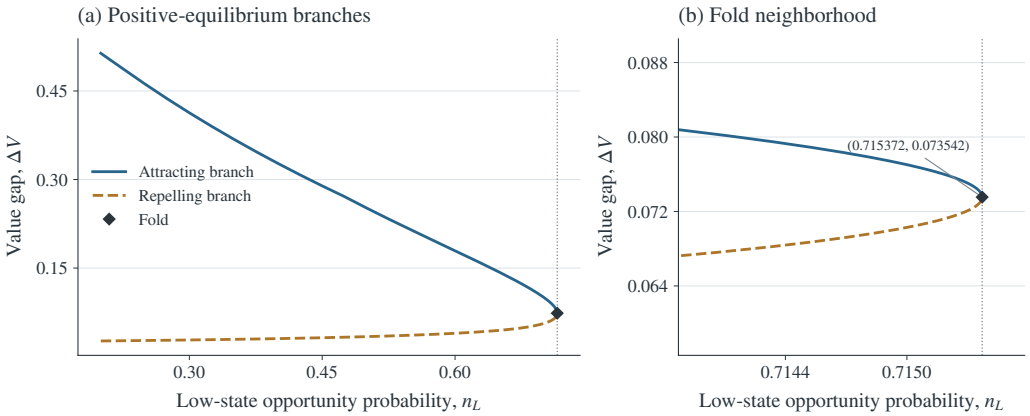


Figure 7: Exposure branches as the low-state opportunity probability rises. The right panel enlarges the fold neighborhood. The diamond solves the fold equations; line styles use the conjecture-iteration classification in Figure 6.

the two rates to 0.01 and 0.99, at a fixed mean of one half, raises that gap to 0.5765 and the group credibility gap to 0.2883. Conditional population-weighted adoption remains close to 0.2496, while the value gap falls from 0.2174 to 0.2104. Figure 8 places the allocation change beside the implied distortion in the public record.

Aggregate adoption can conceal a substantial redistribution of compute. The nearly unchanged purchase rate in this illustration coexists with a widening quality difference across requests. The public sample increasingly emphasizes the requests toward which the provider directs effort.

6. Market design and scope

6.1. Auditing hidden effort

An effort audit is useful when its signal distinguishes productive actions. For an audit outcome J to carry information about high versus low compute, it must satisfy

$$\Pr(J \mid G) \neq \Pr(J \mid P). \tag{56}$$

The common release time in Equation (3) contains no such distinction. A trusted record of productive completion and public release can separate the two clocks. Its incentive value depends on the resulting detection gap Δ_A.

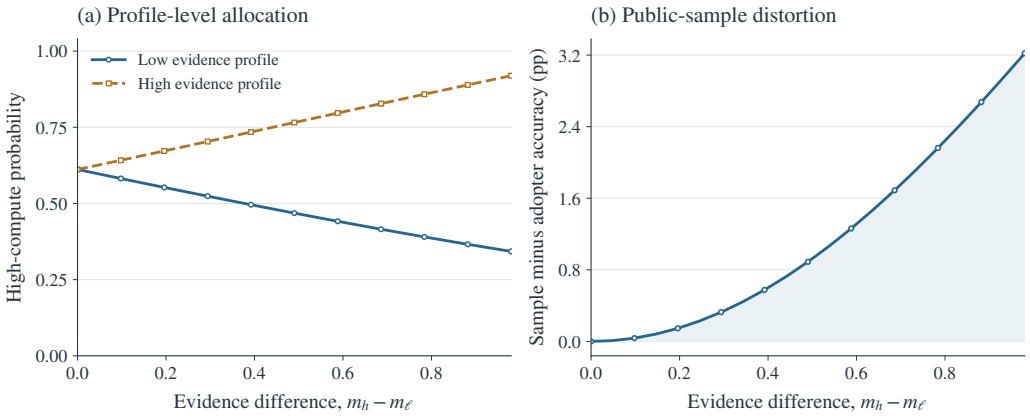


Figure 8: Evidence heterogeneity at a fixed mean verification probability. Panel (a) separates the two profile-level compute probabilities. Panel (b) reports the gap $100(S_0 - \bar{p})$ in percentage points, which incorporates adoption-selected evidence weights. All 21 path points enter the curves.

Different checks reveal different parts of the production process. Correctness is informative because $\Delta p > 0$, although an individual outcome remains noisy. Token counts measure quantity, while model-identity checks target substitution across models. Statistical detection and verifiable computation pursue these margins through different technical means (Cai et al., 2025; Sun et al., 2025a; Wang et al., 2025; Guo et al., 2026). A check enters this model through (α_G, α_P, a) when its action-conditional signal probabilities and coverage can be specified.

Audit targeting changes the distribution of effort across requests. Equation (46) gives a larger incentive increment to requests with less natural evidence. The auditor can therefore reduce a hidden quality gap by targeting missing evidence, even without observing the private profile. Audit accuracy also affects profits and exposure transitions: a higher favorable-signal probability under high compute can have different consequences from a lower probability under low compute, even when both changes produce the same Δ_A.

6.2. Making public evidence representative

Representative evidence requires coverage beyond naturally visible requests. The covariance identity applies even when every recorded outcome is accurate. A designer concerned with the public quality record must therefore examine

the inclusion rule alongside the reliability of its labels.

Independent sampling extends coverage beyond the naturally observed set of requests. It requires no access to the provider's private score. Broader coverage weakens the link between inclusion and correctness at a fixed service schedule, with the reduction in the unweighted sample gap given by Proposition 4.

Certification has different meanings for effort and measured performance. A compute-integrity statement concerns the hidden productive action. A representativeness statement concerns the coverage of observed outcomes. A corrected quality estimate uses the request roster and inclusion probabilities, as in Equation (45). These distinctions identify what a public claim can establish: effort-audit signals support the exposure rule, while reliable correctness labels support outcome statistics.

Outcome-dependent verification makes the current exposure state relevant to effort. Set audits and transfers to zero, and let success and failure be observed with probabilities ν_z^1 and ν_z^0. The resulting dynamic cutoffs are

$$\mathfrak{D}_z^H = \delta \Delta p \, \nu_z^0 \Delta V, \qquad \mathfrak{D}_z^L = \delta \Delta p \, \nu_z^1 \Delta V. \tag{57}$$

High exposure gives the provider an incentive to avoid public failure. Low exposure rewards a public success. Their equality in the baseline follows from $\nu_z^0 = \nu_z^1 = m_z$; unequal observation rates require state-specific policies. The covariance identity continues to describe the record. Similarly, full high compute removes baseline sample bias because it gives all requests a common success probability. Task-specific accuracy technologies can preserve composition effects even under a common compute allocation.

6.3. Scope and limitations

The model isolates verification incentives within a fixed service environment. A single provider faces an exogenous Fast benchmark, with posted prices and an external two-state exposure rule. The equilibrium class uses a common inner solution across states. Capacity constraints and rival price choices are outside this environment. The numerical economy supplies a theoretical illustration without empirical calibration. Its attraction labels describe conjecture iteration; behavioral selection or different inner solutions across states would require an enlarged equilibrium system.

The verification comparisons depend on feasible measurement of the relevant object. Productive-compute audits require trustworthy execution

records, and random outcome attestation requires recoverable ground truth for sampled requests. The payoff system excludes privacy losses and verification resource costs. The comparisons consequently identify incentive and sampling effects; a welfare ranking would also need these costs.

7. Conclusion

Control over answer release allows hidden quality to vary within one service category. A provider can meet the same posted latency with different amounts of productive compute. The resulting accuracy difference becomes economically consequential when the provider privately predicts which requests will generate public evidence.

Public exposure gives these verification forecasts incentive value. Requests with stronger evidence consequences can receive more effort because their outcomes put more future trade at stake. Adoption changes the distribution of those requests, and natural verification selects from the quality schedule that the provider has chosen. The public-sample gap records this connection through the covariance between inclusion and correctness.

Verification can change either effort allocation or outcome coverage. Missing-evidence audits lower the direct incentive slope toward naturally visible requests. The equilibrium illustrations distinguish full high-compute saturation from interior reversal of the service ordering. Independent attestation instead expands outcome coverage at a fixed quality schedule. With known inclusion probabilities and a request roster, partial coverage also supports unbiased quality estimation.

Future trade sustains the return to hidden compute. The numerical price and exposure folds show how that return can disappear as adoption contracts or the exposure states become too similar. The model's central implication is therefore about the connection between service incentives and the public record: a provider's ability to predict verification affects both the quality users receive and the evidence available to assess it.

Appendix A. Proofs

Appendix A.1. Hidden-action cutoff

Current adoption is determined before the provider observes z and c. Actions G and P produce the same displayed release time. Subtracting the

payoff under P from the payoff under G removes current fixed-price revenue and the common low-compute cost. The remaining payoff difference is

$$\begin{aligned}\zeta_x \Delta x + m_z \zeta_Y \Delta p + (1 - m_z) a \Delta_A F \\ + \delta[m_z \Delta p + (1 - m_z) a \Delta_A] \Delta V - c.\end{aligned} \tag{A.1}$$

The terms before $-c$ equal $\mathfrak{D}_z$. The difference is strictly decreasing in c. High compute is optimal exactly when $c \leq \mathfrak{D}_z$, which proves Equations (15) and (16). □

The expected margin in Equation (18) follows by averaging over the cutoff rule. Expected productive compute is $x_L + g_z \Delta x$. Expected correctness is $p_L + g_z \Delta p$. The expected adverse-audit probability for a request without natural evidence is

$$(1 - g_z)(1 - \alpha_P) + g_z(1 - \alpha_G) = 1 - \alpha_P - g_z \Delta_A. \tag{A.2}$$

The unconditional incremental high-compute cost is K_z from Equation (17). Substitution gives M_z. □

Appendix A.2. State invariance and the equilibrium correspondence

Fix a conjectured ΔV. The discipline index depends on the profile posterior and the value gap. The opportunity factor n_s enters after conditional adoption and compute decisions. Multiplying every group-profile weight by n_s leaves Equation (6) unchanged. Thus the vector (m_z, g_z, d_i) is common across exposure states for a fixed inner solution.

Aggregating group-profile signal probabilities gives Equations (22) and (23). The same factorization gives $\pi_s = n_s \bar{\pi}$. Subtracting the two Bellman equations yields Equation (27).

For the compact bound, use $|\pi_H - \pi_L| \leq |n_H - n_L| \bar{\pi}^{\text{abs}}$. Since $\ell_H + u_L \geq 0$,

$$|\mathcal{H}(V, m)| \leq \frac{|n_H - n_L| \bar{\pi}^{\text{abs}}}{1 - \delta + \delta(\ell_H + u_L)} \leq \bar{B}_V. \tag{A.3}$$

Every value gap in the common-inner-solution class lies in the stated interval. If the composite map $h_*(V)$ is continuous on a closed interval, opposite signs of $h_*(V) - V$ at the endpoints give a fixed point by the intermediate value theorem. State-dependent selections can require separate conditional margins and an enlarged equilibrium system. □

Appendix A.3. Existence of a two-group inner solution

Fix $V > 0$. Suppose both group masses and every $Q(z \mid i)$ are positive. Let H_c be continuous and suppose the induced demand of each group is positive for every posterior in $[m_\ell, m_h]$. For an adopter share $q \in [0,1]$ of group h, set

$$m_z(q) = \frac{(1-q)Q(z \mid \ell)m_\ell + qQ(z \mid h)m_h}{(1-q)Q(z \mid \ell) + qQ(z \mid h)}. \tag{A.4}$$

Use the cutoff to obtain $g_z(q)$, then group credibility and demand $d_i(q)$. These objects are continuous under Equation (11). The map

$$T_V(q) = \frac{\mu_h d_h(q)}{\mu_\ell d_\ell(q) + \mu_h d_h(q)} \tag{A.5}$$

is continuous from $[0,1]$ into $(0,1)$. Thus $T_V(0) > 0$ and $T_V(1) - 1 < 0$, so an inner fixed point exists. These conditions hold for the uniform-cost unaudited benchmark when $m_\ell > 0$: every cutoff is at least $\delta \Delta p V m_\ell > 0$, which gives positive credibility and strictly positive exponential-tail demand. This establishes inner existence at each positive conjecture; it does not establish uniqueness or a continuous selection across conjectures. □

Appendix A.4. Zero-demand beliefs and continuity of value

Set $a = \zeta_x = \zeta_Y = 0$, with $H_c(0) = 0$ and H_c continuous at zero. For any admissible posterior vector and $V \geq 0$,

$$0 \leq g_z \leq H_c(\delta \Delta p V), \qquad 0 \leq \chi_i \leq H_c(\delta \Delta p V). \tag{A.6}$$

Hence allocations vanish uniformly as $V \downarrow 0$. Since $P_D - P_F > 0$, Equation (10) gives

$$d_i \leq \exp\left[-\frac{\gamma_i(P_D - P_F)}{w_i \Delta p H_c(\delta \Delta p V)}\right] \longrightarrow 0, \tag{A.7}$$

with zero demand when the CDF bound is zero. Margins are bounded, so $|\bar{\pi}| \leq M \sum_i \mu_i d_i \to 0$ for a finite M. The denominator of Equation (27) is at least $1 - \delta > 0$. Therefore $\mathcal{H}(V, m) \to 0$ uniformly over inner solutions. Defining $h_*(0) = 0$ makes the composite value map continuous at zero, independently of the off-path posterior convention. With uniform costs, the displayed demand bound also implies $h_*(V)/V \to 0$. □

Appendix A.5. Proof of Proposition 1

For any x satisfying $h(x) \leq \tau$, choose $d(x) = \tau - h(x)$. This choice is feasible and gives displayed latency $T = \tau$. Consider a candidate high-compute action x_H and a lower action x_L. The private gain from the lower action at the same display is

$$C(x_H, c) - C(x_L, c) - [\psi(\tau - h(x_L)) - \psi(\tau - h(x_H))] - \mathfrak{D}_z. \tag{A.8}$$

Condition (32) makes this gain positive. The deviation preserves the observed signal, so the user cannot respond at the point of service. If padding is infeasible, equal latency requires $h(x_L) = h(x_H)$. Strict monotonicity of h then implies $x_L = x_H$. □

Appendix A.6. Adoption-selected posterior gap

Consider two equal-mass groups. Let profile z_h identify the high-verifiability group with probability ρ, and let z_ℓ identify the low-verifiability group with the same probability. Bayes' rule among Deep adopters gives

$$m_{z_h} = \frac{d_\ell(1-\rho)m_\ell + d_h\rho m_h}{d_\ell(1-\rho) + d_h\rho}, \tag{A.9}$$

$$m_{z_\ell} = \frac{d_\ell\rho m_\ell + d_h(1-\rho)m_h}{d_\ell\rho + d_h(1-\rho)}. \tag{A.10}$$

Putting the fractions over a common denominator and collecting terms gives Equation (35). Expanding the denominator gives Equation (36), which is weakly larger than $d_\ell d_h$. The absolute selected gap is therefore weakly smaller than the equal-demand benchmark. □

Appendix A.7. Proof of Lemma 1

Because E is binary,

$$S_0 = \frac{\mathbb{E}[EY]}{\mathbb{E}[E]}. \tag{A.11}$$

The covariance definition gives

$$\mathbb{E}[EY] = \mathbb{E}[E]\mathbb{E}[Y] + \mathrm{Cov}(E, Y). \tag{A.12}$$

Divide by $\mathbb{E}[E]$ and subtract $\bar{p} = \mathbb{E}[Y]$ to obtain Equation (38).

Apply the law of total covariance conditional on z:

$$\mathrm{Cov}(E, Y) = \mathrm{Cov}(\mathbb{E}[E \mid z], \mathbb{E}[Y \mid z]) + \mathbb{E}[\mathrm{Cov}(E, Y \mid z)]. \tag{A.13}$$

Substitute $m(z) = \mathbb{E}[E \mid z]$ and $p(z) = \mathbb{E}[Y \mid z]$. Then use $\Pr(E = 1) = \mathbb{E}[m(z)]$. This yields Equation (39). □

For Equation (40), let the adopter group be $I \in \{\ell, h\}$. Conditional independence within group gives $\mathbb{E}[EY \mid I = j] = m_j p_j$. Binary covariance algebra gives

$$\mathrm{Cov}(E, Y) = \omega_\ell \omega_h (m_h - m_\ell)(p_h - p_\ell). \tag{A.14}$$

Under the symmetric profile,

$$p_h - p_\ell = (2\rho - 1)(g_{z_h} - g_{z_\ell})\Delta p. \tag{A.15}$$

Substitution proves the stated expression and sign. □

Appendix A.8. Proof of Proposition 3

Conditional independence gives $\mathbb{E}[Y \mid z] = p_L + \Delta p H_c(A + B m_z)$ and zero within-profile covariance. Substituting in Lemma 1 yields Equation (41). Let M, M' be independent draws from the adopter distribution of m_z, and write $f(m) = H_c(A + Bm)$. The identity

$$2\,\mathrm{Cov}(M, f(M)) = \mathbb{E}[(M - M')(f(M) - f(M'))] \tag{A.16}$$

gives the weak sign of B, because H_c is nondecreasing. For $B \neq 0$, the integrand has a strict common sign on a set of positive probability exactly when $f(M)$ is nonconstant. Uniform costs and interior actions imply $f(m) = (A + Bm)/\bar{c}$. Thus $\mathrm{Cov}(M, f(M)) = B\,\mathrm{Var}(M)/\bar{c}$, which yields Equation (42). □

Appendix A.9. Proof of Proposition 4

Independence of R gives

$$\mathbb{E}[E_r] = r + (1 - r)\Pr(E = 1), \tag{A.17}$$
$$\mathbb{E}[E_r Y] = r\mathbb{E}[Y] + (1 - r)\mathbb{E}[EY]. \tag{A.18}$$

Subtracting $\mathbb{E}[E_r]\mathbb{E}[Y]$ yields

$$\mathrm{Cov}(E_r, Y) = (1 - r)\,\mathrm{Cov}(E, Y). \tag{A.19}$$

Apply Equation (38) to E_r to obtain Equation (44). For nonzero covariance,

$$\frac{d}{dr}|S_r - \bar{p}| = -\frac{|\,\mathrm{Cov}(E, Y)|}{[r + (1 - r)p_E]^2} < 0. \tag{A.20}$$

At $r = 1$, every request enters the outcome sample and the gap is zero. □

For Equation (45), condition on the complete roster $\mathcal{R} = \{(E_j, Y_j)\}_{j=1}^{N}$. Each draw satisfies $\mathbb{E}_R[R_j/r] = 1$. Therefore

$$\mathbb{E}_R[\widehat{p}_r \mid \mathcal{R}] = \frac{1}{N}\sum_j Y_j, \qquad \mathrm{Var}_R(\widehat{p}_r \mid \mathcal{R}) = \frac{1-r}{N^2 r}\sum_j (1-E_j)Y_j. \tag{A.21}$$

The variance follows from independent draws across requests and $Y_j^2 = Y_j$. Only outcomes missing from the natural record contribute sampling variance. □

Appendix A.10. Proof of Proposition 5

Equation (33) can be written as

$$B(a, \Delta V) = \Delta p(\zeta_Y + \delta \Delta V) - a\Delta_A(F + \delta \Delta V). \tag{A.22}$$

Under the stated condition, $B(a, \Delta V)$ is strictly decreasing. Solving $B(a, \Delta V) = 0$ gives Equation (48). Since H_c is increasing, the interior ordering of high-compute probabilities follows the sign of $B(a, \Delta V)$. Boundary clipping can replace a strict gap with equality. □

Appendix A.11. Local fold condition

On the C^2 inner selection in Section 4.6, Taylor expansion at (V^*, P_D^*) gives $F^{\mathrm{fp}} = F^{\mathrm{fp}}_{P_D}(P_D - P_D^*) + \frac{1}{2}F^{\mathrm{fp}}_{VV}(V - V^*)^2 + o(|P_D - P_D^*| + |V - V^*|^2)$. The nonzero derivatives in Equation (53) imply two nearby roots on one side of the critical parameter and none on the other, within this local selection. □

Appendix B. Numerical implementation and additional figures

The numerical routines solve buyer composition before exposure-value consistency. The inner posterior solver starts from Equation (29), evaluates demand ratios in the log domain and uses bounded least squares as a fallback. The outer routine scans the compact interval in Equation (30), with refinement at sign changes and checks for near-tangencies.

Buyer composition becomes concentrated as trade approaches zero. Table B.4 compares the positive-demand posteriors with the zero-trade convention. The remaining buyers increasingly come from the group receiving higher expected quality. The value map's continuity follows separately from the uniform bound proved in Appendix A.4.

Table B.4: Posterior composition near zero demand. The zero row uses the off-path convention; positive rows solve the adoption-posterior system.

Value conjecture V	m_{z_ℓ}	m_{z_h}
0	0.3500	0.6500
10^{-3}	0.7140	0.7890
10^{-4}	0.7844	0.7982
10^{-5}	0.7978	0.7997

The phase tables describe detection under a specified algorithm. Their binary indicator is one when an attracting branch with $\Delta V > 10^{-8}$ is detected and zero when no such branch is detected; solver failures have a separate missing status. This resolution rule leaves extremely small positive branches outside the indicator. At $P_D = 0.45$ and $a = 0.01$, the mapping gives $\mathcal{H}(0) \simeq 1.388 \times 10^{-294}$. Substitution of this value returns the same floating-point number. This is a near-zero candidate diagnostic. At $a = 0.001$, $\log \mathcal{H}(0) \simeq -6680.43$, which lies below the representable range in ordinary floating-point arithmetic. Positive compute and positive margins at the origin distinguish both cases from the unaudited zero-demand branch.

Holding continuation values fixed exposes the direct effect of audit targeting. The illustration retains the baseline value gap and posteriors, with $\bar{c} = 2\delta\Delta V\Delta p$. This normalization gives $g_z = 1/2$ at $B = 0$ under zero transfers. Figure B.1 traces $a \in [0, 1.25a^\dagger]$ while both probabilities remain interior. Figure 4 provides the corresponding equilibrium illustration with endogenous value and posteriors.

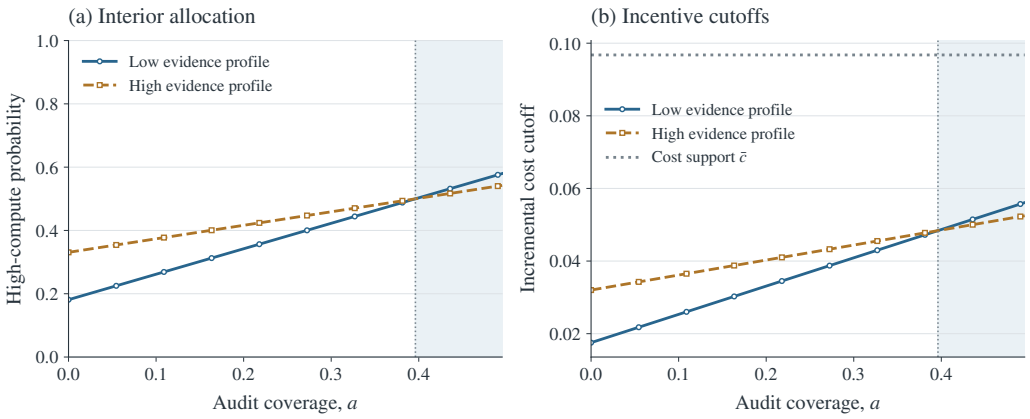


Figure B.1: Conditional sorting reversal at fixed baseline value and posteriors. Panels (a) and (b) compare action probabilities and cutoffs; cost support is $2\delta\Delta V\Delta p$. The vertical guide marks $a^{\dagger}$, and shading indicates the region with $B < 0$.

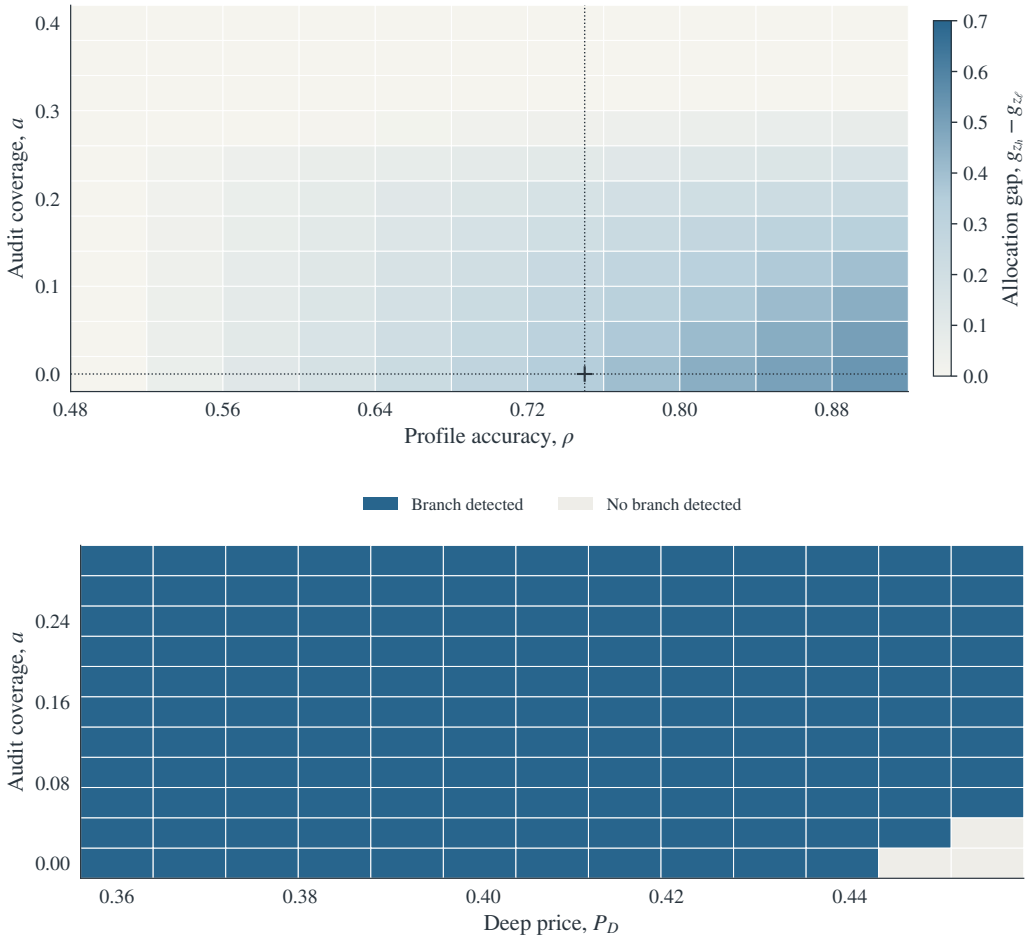


Figure B.2: Audit and price diagrams on the computed grids. The upper map reports the signed score-level allocation gap; dotted guides mark the baseline. The lower map distinguishes detection and non-detection of an attracting branch above $\Delta V = 10^{-8}$ under the stated initialization rule. Each cell represents a computed grid point; boundaries are not interpolated.

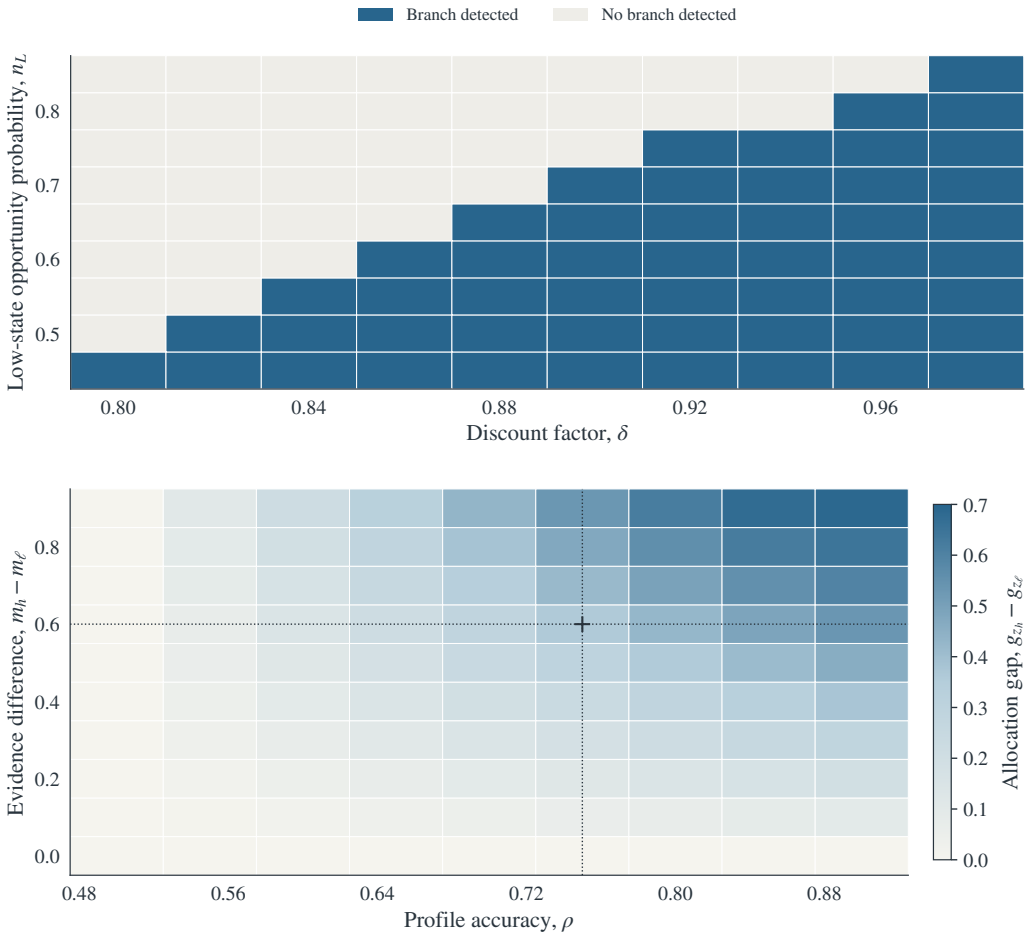


Figure B.3: Exposure and evidence diagrams on the computed grids. The upper map reports attracting-branch detection above $\Delta V = 10^{-8}$. The lower map reports $g_{z_h} - g_{z_\ell}$, with the same color scale as the upper map in Figure B.2; dotted guides mark baseline values. Each cell retains its computed status.

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During the preparation of this work, the author used ChatGPT (OpenAI) for grammar correction, language polishing and LaTeX typesetting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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